

Appian Interview Experience

Pre-requisites - Java

Number of rounds - 4

Online Assessment - Coding round

Technical Round 1 - 45 minutes

Technical Round 2 - 45 minutes

HR Interview - 20 minutes

Online Assessment

There were 5 MCQ Questions and 2 coding questions.

The MCQ questions were debugging, predict the output type questions.

The coding questions (Java was the only language available) were

1. A easy problem involving HashMap. (Leetcode easy level)

2. Breadcrumbs in a website.

Given an `ArrayList<ArrayList<String>>` navigations generate the actual navigation tree.

Example

```
[
    ["Root", "Profile", "Settings"],
    ["Root", "Profile", "Albums"],
    ["Root", "Settings"],
]
```

Helper class `TreeNode` was given

```
class TreeNode {
    String value;
    ArrayList<TreeNode> children;
    void addChild(TreeNode c){
        this.children.add(c);
    }
    //a function to print the tree
}
```

The function that you need to implement is

```
void createTree (ArrayList<ArrayList<String>> paths) {
}
```

The solution is

```
import java.util.*;

class TreeNode{
    String value;
    ArrayList<TreeNode> children;
    TreeNode(String str) {
        this.value = str;
    }
    void addChild (TreeNode c) {
```

```

        this.children.add(c);
    }
}
class answer{
    void createTree(ArrayList<ArrayList<String>> paths){
        TreeNode parent = new TreeNode("Root");
        HashMap<String, TreeNode> map = new HashMap<>();
        for (ArrayList<String> path : paths) {
            for (int i = 1 ; i < path.size(); i++) {
                boolean alreadyIncluded = false;
                String parent = path.get(i-1);
                String current = path.get(i);
                for (TreeNode child : parent.children){
                    if (child.value == current){
                        alreadyIncluded = true;
                        break;
                    }
                }
                if (!alreadyIncluded) {
                    TreeNode child = null;
                    if (!map.containsKey(current)){
                        child = new TreeNode(current);
                        map.put(current, child);
                    }
                    else{
                        child = map.get(current);
                    }
                    parent.addChild(child);
                }
            }
        }
    }
}

```

38 students were shortlisted for the interview.

First Technical Round

1.A coding question (Find the majority element in the array)

<https://leetcode.com/problems/majority-element/>.

You have to solve the problem in constant space and in linear time. Use the Moore's voting algorithm(<https://www.geeksforgeeks.org/boyer-moore-majority-voting-algorithm>). You have to actually code out the solution in a editor and pass the test cases. Then a follow up question on how to find the element that occurs more than 40% of the time in the array was asked. The constraint given was constant space. Sorting the array taking the count would work. Time given for the coding question was 20 minutes. Then resume related questions were asked. As I had mentioned React JS in my skills questions related to React was asked.

1. Advantages of using React

- 2.What is prop drilling.
- 3.UseContext hook.
- 4.What is JSX.
- 5.Advantages of using JSX over traditional JS.

Difficulties faced during the project. I mentioned how deploying the project was difficult as the server cache was less. So they asked about the deployment experience in azure and how to remote sign in and check logs.

2nd Technical Round.

This round again started with a technical question. The technical question was the same breadcrumbs question asked in the OA. A few modification and edge cases in the question was asked. What to do if there were two paths to a particular page and so on. Again you had to code the solution within 20 minutes. Then Resume related questions were asked, this round focused on my intern experience. My intern work was discussed and questions related to it was asked.

3rd HR Round.

The HR Round was like Ask Me Anything session with the HR. There weren't much HR questions asked just normal talk about college.I asked about their work culture, how the company works and about their product.

Finally 4 from CSE were selected for FTE.

Key takeaways.

- 1.Be strong in aptitude(mostly overlooked) and in DSA.
- 2.Know everything that's in your resume.
- 3.Sometimes you may not get shortlisted even if you did the Online Assessment well, it depends on a variety of factors.
- 4.Try to have a CGPA above 9 or 8.5.
- 5.Deploy your projects.